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Computer Networks

Programming Project 3

Introduction to GNS3

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Part 2 Familiarization with GNS3 Topology

1.1 Objective

The goal of this part is to make a simple local area network in GNS3. We set up IP address for one subnet and check the connection between computers and switches.

1.2 Topology Overview

The network has many switches. Switch 3 is the main switch in the middle. It connects to Switch 1 and Switch 2. There are four virtual computers at the end. All devices use the same 192.168.1.0 network address.

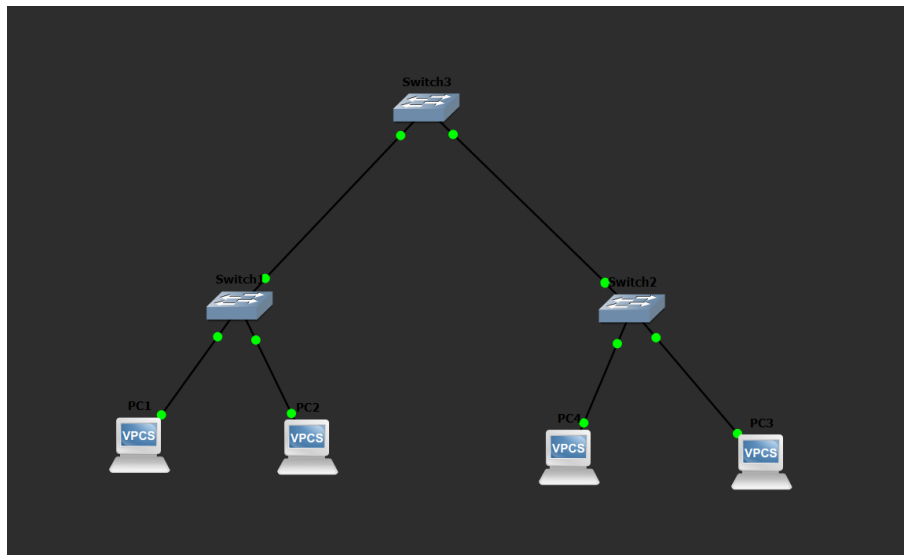


Figure 1: Hierarchical Multi Switch Topology

1.3 Step by Step Implementation

1. **Adding Devices** We put three normal switches in the GNS3 screen. We also put four virtual computers.
2. **Connecting Cables** We connect Switch 3 to Switch 1 and Switch 2. We connect PC1 and PC2 to Switch 1. We connect PC3 and PC4 to Switch 2.
3. **Setting IP Address** We turn on all devices and open the terminal. We give static IP address to all computers using the ip command.

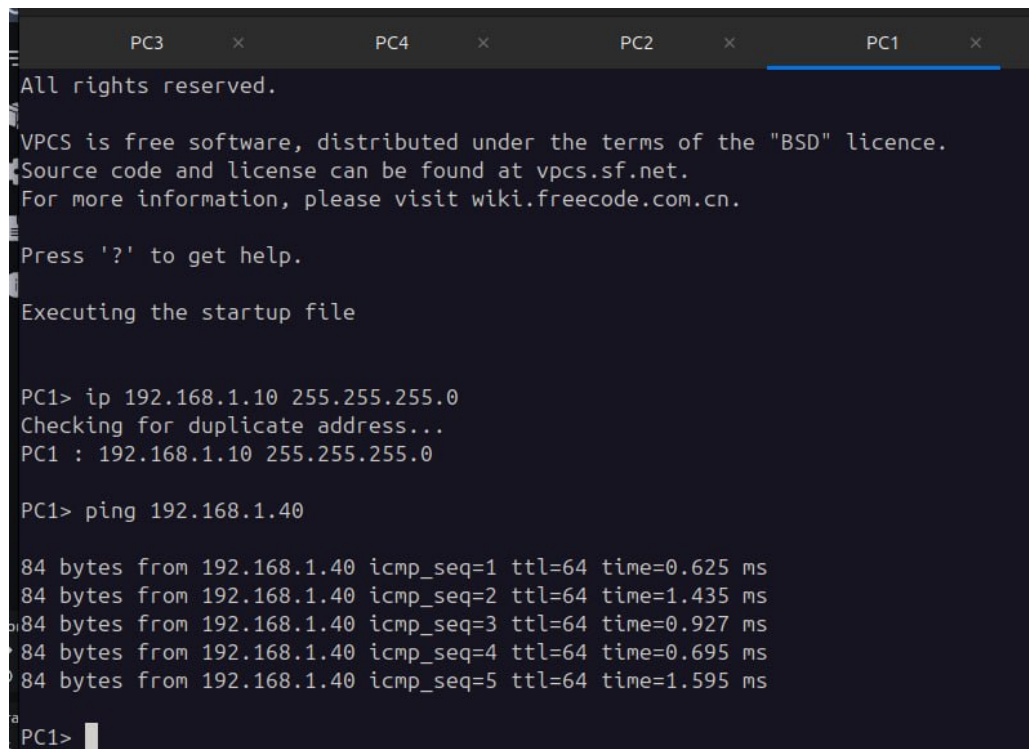
- **PC1** ip 192.168.1.10 255.255.255.0

- **PC2** ip 192.168.1.20 255.255.255.0
- **PC3** ip 192.168.1.30 255.255.255.0
- **PC4** ip 192.168.1.40 255.255.255.0

4. **Testing Connection** We want to see if the network works. We send a ping from PC1 to PC4. We use ping 192.168.1.40 command.

1.4 Results and Analysis

The ping command worked very well. It shows that the switches can find MAC addresses. The packets went from one switch to another switch without problem. The time of the reply was good so the connection is stable.



```
PC3 x PC4 x PC2 x PC1 x
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Press '?' to get help.
Executing the startup file

PC1> ip 192.168.1.10 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0

PC1> ping 192.168.1.40

84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=0.625 ms
84 bytes from 192.168.1.40 icmp_seq=2 ttl=64 time=1.435 ms
84 bytes from 192.168.1.40 icmp_seq=3 ttl=64 time=0.927 ms
84 bytes from 192.168.1.40 icmp_seq=4 ttl=64 time=0.695 ms
84 bytes from 192.168.1.40 icmp_seq=5 ttl=64 time=1.595 ms

PC1> 
```

Figure 2: Successful Ping from PC1 to PC4

Part 3 VLANs and Inter VLAN Routing

2.1 Objective

The goal of this part is to use a router to connect different VLANs. We use a method called Router on a Stick. This lets devices in VLAN 10 talk to devices in VLAN 20 using one physical cable.

2.2 Topology Setup

The network has one router and one switch. We have four virtual computers. The network is cut into two subnets.

- **VLAN 10** Uses network 192.168.10.0 and has PC7 and PC8.
- **VLAN 20** Uses network 192.168.20.0 and has PC9 and PC10.

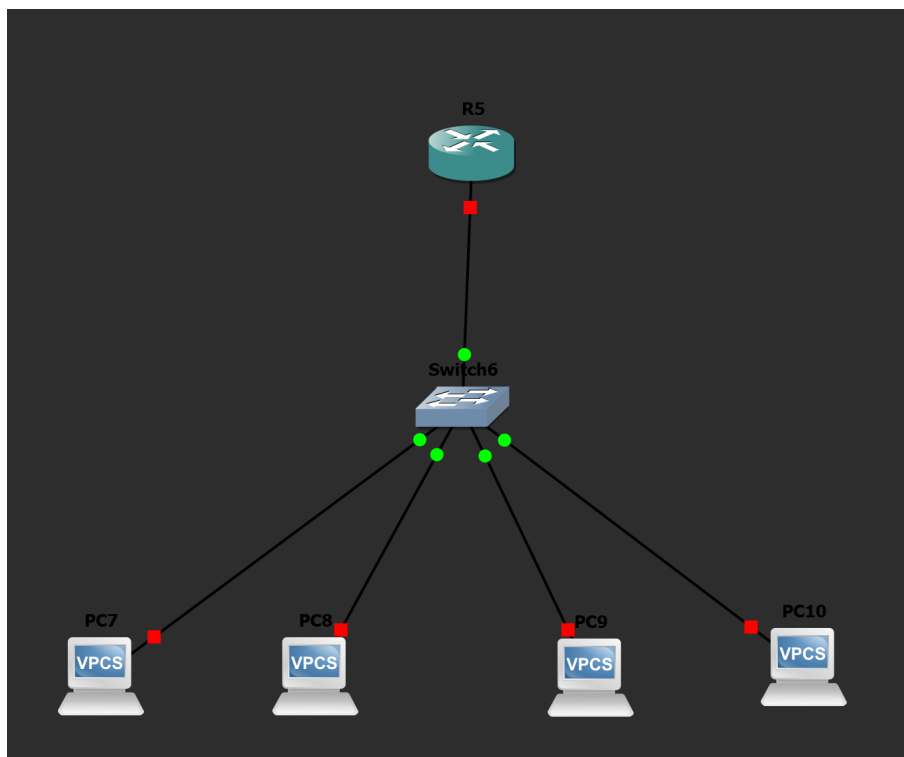


Figure 3: Router on a Stick Topology

2.3 Device Configuration

We configure the switch to have access ports for the computers. We configure the port connected to the router as a trunk port. We create two sub interfaces on the router. We give each sub interface an IP address to act as the gateway for the VLAN. We assign the correct gateway IP to every computer. The screenshot for PC7 shows we typed `ip 192.168.10.10 255.255.255.0 192.168.10.1`. The last IP is the default gateway. This gateway is the sub interface on the router.

2.4 Verification and Testing

We check the routing table on the router using `show ip route`. The screenshot shows the letter C for connected networks. It shows the router is directly connected to 192.168.10.0 on FastEthernet0/0.10. It also shows it is directly connected to 192.168.20.0 on FastEthernet0/0.20. This proves the virtual sub interfaces work.

```
R5(Config)#end
R5#
R5#
R5#
*Jun 18 10:35:18.095: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, FastEthernet0/0.10
L       192.168.10.1/32 is directly connected, FastEthernet0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, FastEthernet0/0.20
L       192.168.20.1/32 is directly connected, FastEthernet0/0.20
R5#
```

Figure 4: Routing Table on R5

We test the network by sending a ping from PC7 to PC10. The screenshot shows the ping 192.168.20.20 command and the 84 bytes replies. The ping goes from VLAN 10 to the gateway on the router. The router sends it to VLAN 20. The reply comes back without any problem.

```

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For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC7> ip 192.168.10.10 255.255.255.0 192.168.10.1
Checking for duplicate address...
PC7 : 192.168.10.10 255.255.255.0 gateway 192.168.10.1

PC7> ping 192.168.20.20

84 bytes from 192.168.20.20 icmp_seq=1 ttl=63 time=51.446 ms
84 bytes from 192.168.20.20 icmp_seq=2 ttl=63 time=19.239 ms
84 bytes from 192.168.20.20 icmp_seq=3 ttl=63 time=12.699 ms
84 bytes from 192.168.20.20 icmp_seq=4 ttl=63 time=11.845 ms
84 bytes from 192.168.20.20 icmp_seq=5 ttl=63 time=20.817 ms

PC7>

```

Figure 5: Successful Ping from PC7 to PC10

2.5 Bonus Wireshark Packet Analysis

We use Wireshark to look at the trunk link between the switch and the router. The screenshot shows the ICMP ping packets. You can see the source is 192.168.10.10 and the destination is 192.168.20.20. The switch adds an 802.1Q tag to the packets. The tag is hidden in the bottom menu of the Wireshark program but we know it is there. This tag tells the router which VLAN the packet came from. The router changes the tag and sends it to the other VLAN.

57.360.391807	ca:01:5f:4c:00:00	ca:01:5f:4c:00:00	LOOP	60 Reply
58.364.142614	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
59.364.140708	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
60.364.147109	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
61.364.150908	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
62.365.150907	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
63.365.150289	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
64.365.150908	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
65.365.169603	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
66.366.171809	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
67.366.172106	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
68.366.173608	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
69.366.183338	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
70.367.185268	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
71.367.188364	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
72.367.189136	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
73.367.190575	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
74.368.199719	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
75.368.200213	192.168.10.10	192.168.20.20	ICMP	102 Echo (ping) request
76.368.201673	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
77.368.211889	192.168.20.20	192.168.10.10	ICMP	102 Echo (ping) reply
78.370.397233	ca:01:5f:4c:00:00	ca:01:5f:4c:00:00	LOOP	60 Reply

Figure 6: Wireshark Capture of Inter VLAN Traffic

Part 4 Static Routing and Network Redundancy

3.1 Objective

The goal is to use static routing to connect routers. We also make a backup path. If the main path breaks the network uses the backup path.

3.2 Topology Setup

We use four routers and two switches. The network has a diamond shape. R1 is top. R2 is bottom. R3 is left. R4 is right. The main path goes through R1. The backup path goes through R2.

3.3 Primary Configuration

We tell the routers where to send data. We add a primary route. We also add a backup route with a high cost. The router hides the backup route until the primary route fails. The first screenshot shows the trace going through R1.

```
PC5> ping 192.168.4.10

84 bytes from 192.168.4.10 icmp_seq=1 ttl=61 time=145.884 ms
84 bytes from 192.168.4.10 icmp_seq=2 ttl=61 time=36.178 ms
84 bytes from 192.168.4.10 icmp_seq=3 ttl=61 time=50.651 ms
84 bytes from 192.168.4.10 icmp_seq=4 ttl=61 time=48.476 ms
84 bytes from 192.168.4.10 icmp_seq=5 ttl=61 time=34.551 ms

PC5> trace 192.168.4.10
Trace to 192.168.4.10, 8 hops max, press Ctrl+C to stop
 1  192.168.1.1    67.218 ms  10.283 ms  20.179 ms
 2  10.0.0.1      59.278 ms  29.767 ms  30.313 ms
 3  10.0.0.6      83.106 ms  30.686 ms  29.601 ms
 4  *192.168.4.10  41.754 ms (ICMP type:3, code:3, Destination port unreachable)

PC5>
```

Figure 7: Successful Trace Through Primary Path R1

3.4 Manual Forwarding Tables

The assignment requires us to make the static forwarding tables for all routers. Here are the tables we configured.

Router 3 Forwarding Table

- Destination 192.168.4.0 Next Hop 10.0.0.1 Cost 1 Main Path

- Destination 192.168.4.0 Next Hop 10.0.0.9 Cost 10 Backup Path

Router 4 Forwarding Table

- Destination 192.168.1.0 Next Hop 10.0.0.5 Cost 1 Main Path
- Destination 192.168.1.0 Next Hop 10.0.0.13 Cost 10 Backup Path

Router 1 Forwarding Table

- Destination 192.168.1.0 Next Hop 10.0.0.2 Cost 1
- Destination 192.168.4.0 Next Hop 10.0.0.6 Cost 1

Router 2 Forwarding Table

- Destination 192.168.1.0 Next Hop 10.0.0.10 Cost 1
- Destination 192.168.4.0 Next Hop 10.0.0.14 Cost 1

3.5 Link Failure and Asymmetric Routing

We turn off the link between R3 and R1. We trace the path again. The packet goes to R3. R3 uses the backup path to R2. R2 sends it to R4. But R4 still thinks the main path to R1 is working. R4 sends the reply to R1. R1 drops the packet. The second screenshot shows the trace stopping at R2 with stars.

```
PC5> trace 192.168.4.10
trace to 192.168.4.10, 8 hops max, press Ctrl+C to stop
 1  192.168.1.1  41.119 ms  9.656 ms  10.049 ms
 2  10.0.0.9   94.334 ms  20.571 ms  21.320 ms
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
```

Figure 8: Trace Timeout from Asymmetric Routing

3.6 Resolution and Failover

We fix this by turning off the link between R1 and R4. Now R4 knows the main path is broken. R4 uses its backup path to R2. The third screenshot shows the trace going through R2 successfully. The backup path works.

```

PC5> trace 192.168.4.10
trace to 192.168.4.10, 8 hops max, press Ctrl+C to stop
 1 192.168.1.1 19.384 ms 10.027 ms 9.445 ms
 2 10.0.0.9 33.692 ms 20.996 ms 19.331 ms
 3 10.0.0.14 60.956 ms 32.195 ms 38.981 ms
 4 *192.168.4.10 70.572 ms (ICMP type:3, code:3, Destination port unreachable)
PC5>

```

Figure 9: Successful Trace Through Backup Path R2

3.7 Bonus Wireshark Packet Analysis

We put Wireshark on the backup serial link between R3 and R2. We want to analyze the packet table to prove the floating static route works.

Look at the screenshot of the Wireshark table. There are important columns to analyze.

- **Source and Destination** We see the IP addresses 192.168.1.10 and 192.168.4.10 in these columns. These belong to PC5 and PC6.
- **Protocol** We see ICMP in this column. This tells us these are the ping messages.
- **Info** We see Echo ping request and Echo ping reply. This shows the full two way conversation.

In a normal situation with the primary link working this backup table would be empty. The primary link would take all the PC traffic. The fact that we capture these specific PC IP addresses and ICMP packets on this backup link is the final proof. It proves the routers successfully updated their routing tables. It proves the floating static route took over the traffic when the main path died.

37	140.190292	192.168.1.10	192.168.4.10	ICMP	88 Echo (ping) request	id=0xfaf0, seq=1/256, ttl=63 (reply in 30)
38	148.243708	192.168.4.10	192.168.1.10	ICMP	88 Echo (ping) reply	id=0xfaf0, seq=1/256, ttl=62 (request in 37)
39	149.258156	192.168.1.10	192.168.4.10	ICMP	88 Echo (ping) request	id=0xfbf0, seq=2/512, ttl=63 (reply in 40)
40	149.283584	192.168.4.10	192.168.1.10	ICMP	88 Echo (ping) reply	id=0xfbf0, seq=2/512, ttl=62 (request in 39)
42	150.362030	192.168.1.10	192.168.4.10	ICMP	88 Echo (ping) request	id=0xfcf0, seq=3/768, ttl=63 (reply in 43)
43	150.334066	192.168.4.10	192.168.1.10	ICMP	88 Echo (ping) reply	id=0xfcf0, seq=3/768, ttl=62 (request in 42)
44	151.358281	192.168.1.10	192.168.4.10	ICMP	88 Echo (ping) request	id=0xfdf0, seq=4/1024, ttl=63 (reply in 45)
45	151.381355	192.168.4.10	192.168.1.10	ICMP	88 Echo (ping) reply	id=0xfdf0, seq=4/1024, ttl=62 (request in 44)
46	152.397771	192.168.1.10	192.168.4.10	ICMP	88 Echo (ping) request	id=0xfef0, seq=5/1280, ttl=63 (reply in 47)
47	152.428767	192.168.4.10	192.168.1.10	ICMP	88 Echo (ping) reply	id=0xfef0, seq=5/1280, ttl=62 (request in 46)

Figure 10: Wireshark Packet Table Analysis on Backup Link

Part 5 Dynamic Routing with OSPF

4.1 Objective

The goal of this section is to configure OSPF on a four router network. OSPF is a dynamic routing protocol. It automatically learns the network map. We want to prove that OSPF can find equal paths and also fix broken links automatically.

4.2 OSPF Configuration and Analysis

We configured the IP addresses on all interfaces for the four routers. Then we enabled OSPF process 1 and added the connected networks to Area 0.

After the routers sent their hello messages we analyzed the OSPF tables on Router 1. The neighbor table shows that R1 successfully formed relationships with its neighbors. The state is FULL for both connections on Serial2/0 and Serial2/1.

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.34.1	0	FULL/ -	00:00:30	192.168.14.2	Serial2/1
192.168.23.1	0	FULL/ -	00:00:32	192.168.12.2	Serial2/0

```
R1#
```

Figure 11: OSPF Neighbor Table on R1

The OSPF database shows the raw link state data collected from all routers in Area 0. This proves R1 has a complete map of the network topology.

```
R1#show ip ospf database
```

OSPF Router with ID (192.168.14.1) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
192.168.14.1	192.168.14.1	196	0x80000005	0x008724	5
192.168.23.1	192.168.23.1	222	0x80000003	0x007C8F	4
192.168.34.1	192.168.34.1	197	0x80000003	0x00D10A	4
192.168.34.2	192.168.34.2	212	0x80000003	0x00EC54	5

```
R1#
```

Figure 12: OSPF Database on R1

The routing table shows the calculated best paths. We can see routes marked with the letter O. This proves OSPF injected the routes dynamically. For the destination network 192.168.3.0 OSPF found two equal cost paths. One path goes through 192.168.14.2 and the other goes through 192.168.12.2. The router will load balance the traffic across both cables.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0
O       192.168.3.0/24 [110/129] via 192.168.14.2, 00:01:05, Serial2/1
          [110/129] via 192.168.12.2, 00:01:31, Serial2/0
    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, Serial2/0
L       192.168.12.1/32 is directly connected, Serial2/0
    192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.14.0/24 is directly connected, Serial2/1
L       192.168.14.1/32 is directly connected, Serial2/1
O       192.168.23.0/24 [110/128] via 192.168.12.2, 00:01:41, Serial2/0
O       192.168.34.0/24 [110/128] via 192.168.14.2, 00:01:05, Serial2/1
```

Figure 13: OSPF Routing Table on R1

4.3 Connectivity and Failover Testing

We tested the network by sending a ping from PC1 to PC2. The first test shows complete success with 84 byte replies. This proves the dynamic routes are actively forwarding data.

```
PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> ping 192.168.3.10
84 bytes from 192.168.3.10 icmp_seq=1 ttl=61 time=105.688 ms
84 bytes from 192.168.3.10 icmp_seq=2 ttl=61 time=91.480 ms
84 bytes from 192.168.3.10 icmp_seq=3 ttl=61 time=92.578 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=61 time=90.733 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=61 time=91.141 ms

PC1> █
```

Figure 14: Successful Ping from PC1 to PC2

To test the convergence time of OSPF we deleted one of the active serial cables while the ping was running. The ping command dropped exactly two packets showing a timeout. Then OSPF automatically recalculated the network map and switched to the backup path. The ping recovered immediately. This proves dynamic routing is highly fault tolerant compared to the static routing tested in Part 4.

```
PC1> ping 192.168.3.10
192.168.3.10 icmp_seq=1 timeout
192.168.3.10 icmp_seq=2 timeout
84 bytes from 192.168.3.10 icmp_seq=3 ttl=61 time=94.458 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=61 time=93.543 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=61 time=91.306 ms

PC1> █
```

Figure 15: OSPF Automatic Failover and Recovery